

Lab Session 1

In24-S2-IN1130 - Structured Programming II

Passing Array Elements to Functions

Exercise 1

Display All Elements Using Element Passing

- Write a program to:
- Declare an array of integers
- Use a loop to pass each element to a function
- Print each element inside the function

Exercise 2

Check Even or Odd

- Write a program to:
- Pass each array element to a function
- Check whether the element is even or odd
- Display the result

Exercise 3

Square of Each Element

- Write a program to:
- Pass each array element to a function
- Calculate and print the square of each element

Exercise 4

Find Maximum Using Element Comparison

- Write a program to:
- Pass two array elements at a time to a function
- Compare them and return the larger value
- Determine the maximum element in the array

Passing Arrays to Functions

Exercise 7

Display Array Elements

- Write a program to:
- Declare an integer array of size 5
- Pass the entire array to a function
- Print all elements inside the function

Exercise 8

Sum of Array Elements

- Write a program to:
- Input 6 integers into an array
- Pass the array to a function
- Calculate and return the sum of all elements

Exercise 9

Modify Array Elements

- Write a program to:
- Pass an integer array to a function
- Multiply each element by 2 inside the function
- Display the modified array in main()

Pointers Basic

Exercise 10

Declare and Print Pointer Values

- Write a program to:
- Declare an integer variable with value 10
- Declare a pointer to that variable

Print:

- Value of the variable
- Address of the variable
- Value stored in the pointer
- Value pointed to by the pointer

Exercise 11

Pointer and Data Type

Write a program to:

- Declare variables of type int, float, and char
- Declare pointers for each variable
- Print the value and address of each variable using pointers

Exercise 12

Change Value Using Pointer

- Write a program to:
- Declare an integer variable
- Use a pointer to modify its value
- Print the value before and after modification

Functions (Call by reference)

Exercise 13

Swap Two Numbers

- Write a program to:
- Declare two integer variables
- Swap their values using a function with **call by reference**
- Print values before and after swap

Exercise 14

Multiply Two Numbers

- Write a program to:
- Pass two integer variables by reference to a function
- Multiply them inside the function and store the result in one of the original variables
- Print updated values in main()

Exercise 15

Find Maximum of Two Numbers

- Write a program to:
- Pass two numbers to a function by reference
- Determine the larger number
- Update a third variable in main() with the result

Exercise 16

- **Count Even and Odd Elements**
- Write a program to:
- Pass array elements by reference to a function
- Count number of even and odd elements
- Return counts to main() using reference variables